

Requirements:

LNA/HKA	f_0 (MHz)	$0.9 f_0$ (MHz)	$1.1 f_0$ (MHz)	Gain (dB)	Noise figure (dB)	Input return loss (dB)
LNA	680	646	714	11.58	0.824	8dB

Measurements (use one of the network analyzers for S-parameter measurements. Set the input power level of the network analyzer to -30 dBm for linear measurements. Use the noise figure meter for noise figure measurement. Refer to the “Noise Figure Measurement” document in Moodle.)

Supply voltage (V)	10V
Supply current (mA)	3.46mA
Worst input return loss $- S_{11} _{dB}$ (dB)	8.36 dB (at 646 MHz)
Worst output return loss $- S_{22} _{dB}$ (dB)	14.19 dB (at 714 MHz)
Minimum $ S_{21} _{dB}$ in the band (dB)	10.82 dB (at 714 MHz)
Maximum $ S_{21} _{dB}$ in the band (dB)	11.51 dB (at 646 MHz) (see Figure 1)
Worst noise figure in the band (dB)	(2.14 dB at 714 MHz)

Below pictures are for in case, video is not clear enough and captured better parameter values for the report.

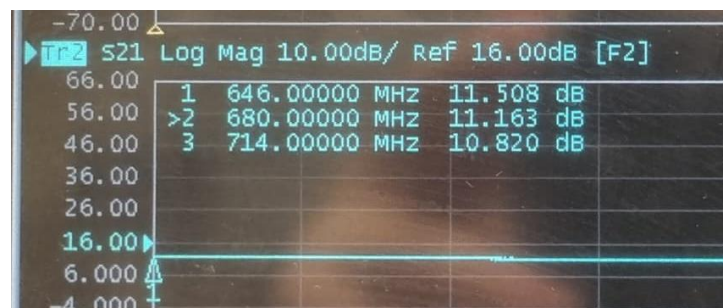


Figure 1: S21 parameter

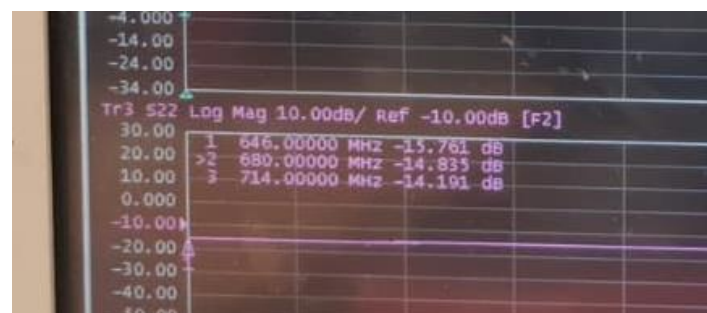


Figure 2: S22 parameter

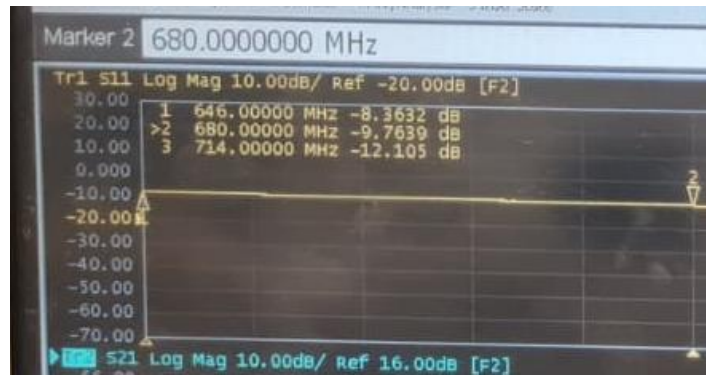


Figure 3: S11 parameter

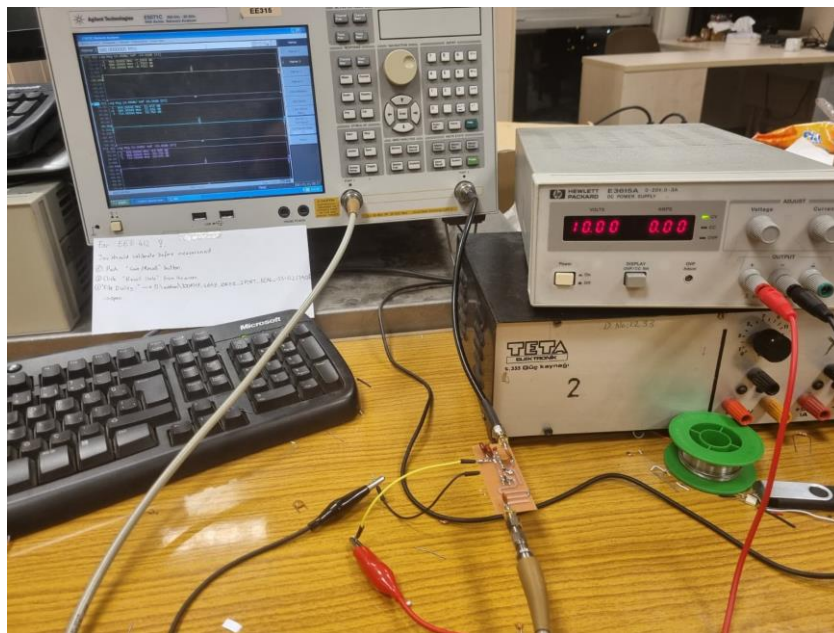


Figure 4: amplifier in the measurement setup, showing the measurement screen

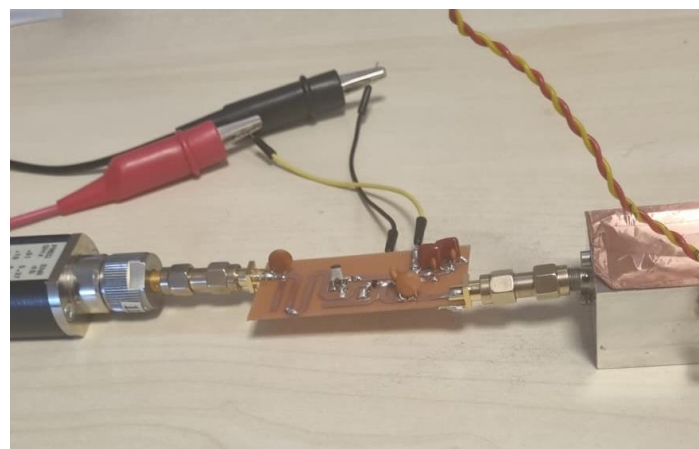


Figure 5: Noise Measurement